

Student: Nele-Rita Pöldma

Internship/Thesis: Combined Internship

Project Title: **Quality Control and Alignment of MRI Data for Investigating Neural Correlates of Empathic Traits**

Understanding the neural basis of empathy is critical for social cognition research. During this internship, a dataset of 248 MRI scans was refined to explore the association between regional gray matter volume and individual differences in empathy using Voxel-Based Morphometry (VBM). The project involved the manual alignment and quality control of structural images, as well as preliminary preprocessing the data. Image alignment, artifact inspection, and data preparation were performed using MATLAB and Statistical Parametric Mapping (SPM). The outputs include a high-quality dataset of aligned MRI scans and quality control documentation. These findings will contribute to a deeper understanding of the structural brain correlates of empathy, with implications for both research and applied fields, such as clinical psychology.

Company/University: Utrecht University

UU Supervisor: Joana B. Vieira, PhD

Name Student: Malte Frerichs

Internship or Thesis: Internship

Title project: Creating a Model to Predict Visual Hemifield and Eye Advantage

(Abstract, 100-150 words)

Binocular rivalry results from competition between both eyes, where perception can alter; it is often also referred to as interocular competition. During binocular rivalry, one eye's image dominates visual perception, leading the other to be suppressed. One facet of interocular competition that has received only brief attention in other papers is the visual hemifield in which the visual stimuli are presented. The goal of this research is to correctly predict which eye and visual hemifield will have the advantage in different situations. It is hypothesized that the eye and visual hemifield that can view the majority of information will hold the advantage. To achieve this, a program was created that can calculate the viewable percentage of a target image for each eye and visual hemifield. This program rests on the assumption that the eye or visual hemifield with more information has the advantage. Moving forward, this program will be tested experimentally from which on improvements to the program as well as the underlying theory can be made.

Name of Company or University: University

(If applicable) External supervisor:

UU Supervisor: Chris Paffen

Name: Zsófia Varga

Internship or Thesis: Thesis

Title project: Effects of stress and cognitive load on frontal P3

Burnout and prolonged stress have been associated with neurophysiological changes, including reduced amplitude of the frontal P3a event-related potential (ERP). This study explored whether everyday stress—without clinical burnout—also relates to diminished frontal P3a amplitude and whether this effect is amplified by cognitive load. Thirty-three participants completed stress-related questionnaires and performed a three-stimulus auditory oddball task before and after playing a multisensory pinball game to induce cognitive load. EEG was used to assess changes in frontal P3 amplitude pre- and post-task. Results will clarify whether everyday stress levels correlate with baseline P3a reductions and whether stress predicts greater post-load decreases in P3a amplitude.

Name company or University: Utrecht University

UU Supervisor: Daan van Rooij, Sam van Bijnen

Name: Simon Punter

Project: Internship

Title: Exploring influence strategies in the context of cognitive warfare

Abstract: This literature review, along with two interviews, investigates influence strategies in the context of cognitive warfare. Using a multi-level framework based on the socio-ecological model, it explores how disinformation and manipulation operate across micro-level cognitive biases, meso-level trust dynamics, and macro-level sociopolitical structures. Findings suggest that interventions such as prebunking, gamified media literacy, and institutional transparency can enhance societal resilience. However, emerging threats, such as digital nudges and opaque algorithmic personalization, increasingly undermine cognitive autonomy. The interviews identified key challenges, including regulatory enforcement gaps, inconsistent terminology, and limited technological capabilities within the EU. Although policy developments, such as the Digital Services Act, provide an initial foundation, effectively mitigating cognitive warfare requires a coordinated approach that combines psychological resilience, social trust, and institutional safeguards. The study concludes that addressing cognitive warfare demands cross-disciplinary collaboration, greater transparency, and a careful balance between innovation and ethical oversight to protect democratic discourse against evolving digital influence strategies.

Company: TNO

External supervisor: Jose Kerstholt

UU supervisor: Krista Overvliet

Name Student: Navya Suneej

Internship or Thesis: Internship

Title project: What role do anticipated outcomes play in behaviour, thereby distinguishing between habits and goal directed behavior?

The research project will be regarding habitual versus goal directed control of behaviour. Specifically, we want to know what happens when people are trained on an instrumental learning task to make specific responses to obtain a reward. According to a dual process model of decision, by repeatedly conducting this behavior a stimulus response habit will form and the anticipation of the outcome becomes increasingly less relevant. Previous studies that have used outcome-devaluation to test whether behavior becomes less driven by its anticipation of the outcome does have some limitations. Therefore, we would like to use a different paradigm using bi-stable perception to obtain an implicit measure of the strength of the outcome anticipation similar to bi-stable spheres.

Name company or University: Utrecht University

(If applicable) External supervisor: Tim van Timmeren

UU Supervisor: Jana Klaus

Name Student: Sarah Vollmer, 1615386

Internship or Thesis: Internship

Title project: Analysing Skin Conductance Responses to Distal and Imminent Threats using Psychophysiological Modelling

This internship project reanalyses skin conductance response (SCR) data from a threat conditioning experiment using Psychophysiological Modelling (PsPM) to compare its efficacy against traditional manual scoring methods. The original study investigated physiological and postural responses to distal and imminent threats, with additional exploration of childhood trauma effects. Participants (N=29) completed a threat-of-shock task featuring three conditions: distal threat, safety, and imminent threat, with occasional unpredictable escape opportunities.

This work aims to explore PsPM's potential to advance threat response research by transforming SCR analysis from descriptive peak-scoring to a more dynamic, model-based approach. To do so the SCR results obtained from the previously mentioned study will be compared to those obtained through the PsPM method. It is expected that these new results will shed light to previously undetected effects, due to PsPM's increased sensitivity.

Name company or University: Utrecht University

UU Supervisor: Joke Baas

name: Richa Zope (4888286)

project: Thesis

title: Exploring the behavioral effects of sensorimotor integration in Neuralgic Amyotrophy with the Rubber Hand Illusion

Neuralgic amyotrophy is a common peripheral nerve disorder that is caused by an auto-immune inflammation of the brachial plexus. Despite peripheral recovery, patients often have residual motor dysfunction leading to complaints such as pain and fatigue due to scapular instability. Neuralgic amyotrophy patients often develop alternative motor strategies as a response to the residual complaints. Previous research suggests NA patients have altered sensorimotor representations related to the affected limb. Maladaptive neuroplasticity in two higher-order visuo-motor brain regions, involved in sensorimotor integration, is suggested to play a role in residual motor dysfunction and pain. The aim of this research is to assess how sensorimotor integration is different in NA patients between the affected and non-affected limb. Additionally, the relationship between sensorimotor integration and upper extremity function will be explored. Treated and untreated NA patients and healthy controls will undergo the Rubber Hand Illusion, a famous paradigm that studies sensorimotor integration in terms of the behavioral outcome measure proprioceptive drift.

Utrecht University

UU supervisor: Renee Lustenhouwer

Name Student: Youssra Abdellaoui

Internship or Thesis: Internship

Title project: Developing and Testing a Coding Scheme for the Annotation of Parental Gestures

This internship focused on developing a reliable method for annotating parental gestures during parent-infant interactions. Research has focused mostly on infants' gesture use, leaving the role of parental gestures largely unexplored. Using video data from the YOUth study, we created and tested a coding scheme to classify different types of parental gestures, such as pointing, object manipulation, and representational gestures. The pilot study showed high inter-rater reliability (Cohen's Kappa = 0.96), confirming the method's consistency. Throughout the annotation process, we refined definitions and rules to ensure clarity and reduce ambiguity. Results showed that object-related gestures were most common, and that parents differ greatly in how often and how they gesture. The final coding scheme provides a valuable tool for future research and may support the development of AI systems for gesture recognition.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Anika van der Klis

Name: Sanyukta Nair

Project: Internship

The Impact of Rewards on Saccade Selection

Attentional selection can be studied by looking at eye movements. Saccade selection- where one chooses to move their eyes, and hence what one attends to, is driven by several factors (salience, observer goals, etc.). Recent research reveals that there is also an impact of the inherent effort of planning and executing eye movements (saccade costs)- with a preference for more ‘affordable’ saccades. This finding that cost optimizations drive choices, should also imply that rewards will affect these saccades. This research addressed if and how saccade costs and rewards together drive saccade selection through a task where participants freely chose between two saccade targets (of either similar or different reward value) in different directions. It was observed that rewards largely predicted saccade selection: participants chose the higher valued target in all conditions where there were differently valued pairs. Additionally, the amount of reward further modulated this choice. Further analysis demonstrates that saccade costs exert an influence on saccade selection even when rewards are introduced, with participants more willing to tolerate higher costs when presented with the choice between high reward and no reward. In line with recent research, our findings bolster the theory that how one ‘looks’ is a neuroeconomic decision process, as saccade selection is a decision that considers both cost (minimization) and reward (maximization). Such an investigation has the potential to provide a good measure of saccade effort- represented in monetary values as opposed to pupil size.

Utrecht University

Supervisor: Christoph Strauch

Co-Supervisor: Damian Koevoet

Poster Presentation Abstract

Student Name: Matej Aluloski (6752071 ; m.aluloski@students.uu.nl)

Thesis Title: Actions Per Minute (APM) as an Analogue to Cognitive Workload: Measuring Performance Through APM, ADT, DPS & Subjective Self Report Criteria in World of Warcraft

Title Project: Cognitive workload, broadly defined as the mental energy expended during task performance, remains difficult to measure due to lacking objective measurement methods. This study aims to determine whether Actions Per Minute (APM) - the frequency of keyboard and mouse inputs - can be used as a proxy for workload in complex human-computer interaction tasks: playing World of Warcraft's Mythic+ Dungeons. Using a within-subjects design (n=26), participants completed three increasingly difficult gaming sessions while measuring their APM. Their APM was correlated with established metrics of performance, such as Avoidable Damage Taken, Damage Per Second and their answers from the NASA-TLX subjective workload inventory. Results indicated that AMP itself is not an effective workload proxy, correlating strongly with performance metrics ($r = 0.68$), but not with subjective workload ($r = 0.12$). This implies that APM functions primarily as a performance indicator rather than reflecting cognitive workload. Practical & theoretical limitations, as well as venues for further research, are also discussed.

Name of Company / University: Utrecht University

External Supervisor: N/A

UU Supervisor: Jeroen Benjamins

Name Student: Reinhard Heideman

Internship or thesis: Internship

Title Project: Improving the UX of Shello's internal tool

Shellos is a market research company that aims to perform quantitative analyses to answer business questions of other companies. For this, they have developed their own tool to help with this research. This tool turned out to have several UX problems and missing features. With this project the aim was to find these limitations and create a list of recommendations for the tool. This was done by performing a heuristic evaluation, semi-structured interviews with employees and making them fill in a PSSUQ questionnaire. The results were thematically analyzed, and based on the results a high-fidelity mockup was created. This was then shown again to employees to get final feedback. In the end, a list of 42 recommendations was created, which were ranked with the MoSCoW method.

Name company or university: Shellos

External supervisor: Nellie Kirschner-Timmer

UU Supervisor: Daan van Rooij

Name Student: Manon Arns

Internship or Thesis: Internship

Title project: Roodlichtnegatie door fietsers: wat zijn voorspellende factoren?

Abstract:

This study aims to provide insight in factors that are predictive for red light infringement (RLI) among cyclists in the Netherlands. For this research, video data of a Dutch crossroad has been studied and analyzed for the following variables: age, gender, duration of the redlight phase, (effective) waiting time, traffic density, group size, and perception of others violating the redlight. Results of several statistical analyses of the annotated data have shown the factors age, gender, effective waiting time, traffic density and group size to have a significant effect on RLI. Since these results are based off analyses of just one crossroad, this study has a somewhat explorative nature, and more research is needed to draw solid conclusions on predictive factors for RLI.

Key words: red light infringement (RLI), traffic safety, waiting time, traffic density, group size

Name company or University: SWOV

(If applicable) External supervisor: Reinier Jansen

UU Supervisor: Leendert van Maanen

Name Student: Lisa D'hopore

Internship or Thesis: Thesis

Title project: **The Non-Linear Relationship Between AI Accurateness and Trust in AI:**
How Does Seeing an Algorithm Err- More or Less Often- Lead to AI Disuse, and How is this Aversion Different With Human-Advice?

(Abstract, 100-150 words)

Studies show that although people dynamically change their trust to match algorithm performance accuracy, they quickly become aversive to using algorithmic advice when observing it make too many mistakes. In our present image labelling study we investigated how an algorithm's accuracy level moderates trust in this algorithm as well as how this relationship might differ from the relationship between human advice accuracy and trust in this human advice. To operationalise trust, we investigated reaction times and agreement rates to algorithm or human advice that were either 60% or 80% accurate. We found people were slower in AI-advice conditions relative to human-advice conditions, suggesting algorithm aversion regardless of the advice accuracy. We also found across both experiments that participants' performance in image labelling was better in AI conditions than in human conditions, potentially the result of people being more careful and attentive to algorithmic advice. Finally, the effects on agreement rates were inconsistent across experiments, although we found further suggestive evidence of algorithm aversion being strengthened with low reliability (60%) opposed to higher reliability (80%) algorithms. Finally, we discussed some limitations of our experiments, notably cover story comprehension and the difference between reliability conditions which might be too small.

Name company or University: Utrecht University

UU Supervisor: Leendert van Maanen

Name: Sana Yasmin

Project: Thesis

Title: How User Interface Affects AI Discoverability & Adaptability: Insights from Capgemini Employees

Company: Capgemini

Ext. supervisor: Christina Potuijt Veelenturf (christien.veelenturf@capgemini.com)

UU supervisor: Maarten Van Der Smagt (m.j.vandersmagt@uu.nl)

Abstract

Artificial Intelligence (AI) is becoming a regular part of our daily lives, from the recommendations we see on Netflix to voice assistants like Siri and Alexa. Despite these advancements, many employees still feel hesitant or unsure about using AI at work. This research explores why that hesitation exists and the role of visual perception, and how it impacts employees' ability to recognize AI in various interfaces. Furthermore, this study looks at whether this recognition impacts employees' understanding and eventual adoption of AI in the corporate sector.

The study looks at whether people are more open to using AI when it's clearly visible (like when there are prompts or cues) compared to when it works quietly in the background. To explore this, we used a mixed-methods approach: a survey with 113 employees to understand their general views on AI, awareness and perception, and an experiment using Signal Detection Theory (SDT) with 40 participants to see if they can spot when AI is involved in certain tools, and follow-up interviews to dig into their thoughts and feelings about what they saw.

This study aims to understand not just if people are resistant to AI but also if visual perception plays a role. Do they find it too complex or simply aren't aware it's even there? By answering these questions, we hope to offer practical insights that can help companies design AI tools that feel more approachable, transparent, and easier to adopt in everyday work life.

Name student: Veerle Hoogerwerf

Internship

Title project: Nature's Music

This study explores the relationship between human musical appreciation and bird song, focusing on the brain's predictive coding system, or the "Bayesian brain." It investigates whether human preferences for bird songs can be explained by the (un)predictability of their syntactic structure. The objective was to develop a psychoacoustic test to assess human appreciation of birdsongs with varying structures. The test was based on semantic scale judgments, where participants rated songs as "Pleasant" or "Unpleasant." Various song manipulations were made, such as altering syllables and rhythms to increase or decrease predictability. Pilot testing with six participants provided valuable feedback, highlighting preferences for song replay and different test formats. The study recommends further test optimization, a larger song database, and the integration of EEG technology to explore brain activity and the evolution of musicality.

University Utrecht

UU supervisor: Gabriël Beckers

Name Student: Novela Perić

Internship

Title project: Optimising the Administration Process of Coachia with the CoVince Digital Environment

CoVince develops immersive digital tools for professionals, offering AI-driven learning experiences, virtual webinars, and conferences. Recently, they introduced report-making tools in response to the needs of Coachia, an outpatient care organization whose coaches work with clients ranging from teenagers to the elderly. Administrative tasks often reduce valuable client time, so CoVince created two solutions: a post-meeting voice-to-text tool for flexible reporting and an AI-powered meeting summarization tool that generates reports from full recordings. To manage the significant workflow shift, mandatory usability testing was conducted as part of a master's program internship. The first two weeks focused on onboarding, brainstorming, and literature review. The following three weeks involved designing experiments, preparing usability tests, and conducting interviews. After the main usability test with Coachia, the company gained valuable insights. In the final two weeks, additional marketing research and testing were conducted with other organizations, culminating in a comprehensive report and internship deliverable.

Name company or University: CoVince

(If applicable) External supervisor: Richard van Tilborg

UU Supervisor: Samson Chota

Name Student: Toos van Dijken

Internship or Thesis: Thesis

Title project: Touch and the Post-Auricular Reflex: Comparing Self, Social, and Mannequin Touch

Abstract

Touch is a fundamental component of human interaction, which can play a crucial role in emotional regulation, social bonding, and well-being. The aim of this project is to investigate whether social touch elicits a stronger post-auricular reflex (PAR) and higher subjective pleasantness ratings compared to self-touch. A sample of participants ($n=60$), take part in a $2 \times 2 \times 2$ within subjects experimental design, testing eight conditions. Either the participant or the researcher touches the arm of the participant, with either their own hand or a make-up brush. Alternatively, either the participant or the researcher touches a fake arm (control conditions), with either their own hand or a make-up brush. Subjective pleasantness ratings will be given after each condition, and electrodes are placed behind both ears and forehead to measure the PAR. It is predicted that the social touch conditions will elicit a stronger PAR and align with higher subjective pleasantness ratings of the participants.

Name company or University: Utrecht University

UU Supervisor: David Terburg